

Reading Support: Delineation and Systems

Companion to Week 12

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A system boundary selects internal relations

Supports: A system is selected, not simply found

A system is an abstraction in which selected components and relations form a unit for analysis. The boundary determines which variables count as internal states and which enter as environmental inputs.

Different boundaries can describe the same events for different purposes. The scientific question is whether the chosen delineation preserves the relations required by the explanation.

MOLR §6.5

A transform changes representation while preserving a relation

Supports: Transforms can reveal a useful invariant description

Generating functions represent a sequence as coefficients of a power series. A Laplace transform represents a time-dependent function in another domain where combination and change may become easier to analyze.

The new representation gains its relevance from a defined correspondence with the original object. Manipulation in the transformed domain does not interpret itself.

MOLR §6.5

An open system can receive information beyond feedback

Supports: An open system retains an independent environmental input

In an open-loop description, a controller can receive environmental information in addition to feedback about the process response. The model therefore leaves room for changing policy or selecting among alternatives on grounds not fixed by the error signal.

The system remains constrained without becoming isolated from the conditions in which it acts.

MOLR §6.5

A closed loop fixes how discrepancy controls response

Supports: A closed loop fixes the route from discrepancy to adjustment

A closed-loop controller compares a desired response with a measured response and adjusts the process under a fixed policy. Feedback can explain regulation and correction within that policy.

The loop does not itself explain where the goal or policy came from, or how an agent can replace it when circumstances change.

MOLR §6.5

Decision and choice locate freedom at different levels

Supports: Choice and decision are not the same abstraction

A decision selects an action from possibilities delineated by a policy, goal, and representation of the situation. Choice can concern the adoption or revision of those very constraints.

Treating every choice as optimization inside a fixed system conceals the capacity to reorganize the system of relevance itself.

MOLR §6.5

The book's stages answer successively stronger questions

Supports: The course has moved through three layers

Basic mathematics supplies the vocabulary for objects and connections. Formal-language theory classifies and describes string admissibility. Automata supply methods for computing those descriptions. Homomorphisms and system boundaries connect the formal results to linguistic organization.

Each later stage presupposes the earlier distinctions without making them interchangeable.

MOLR Preface; Chs. 1–6

Explanation must return through empirical validation

Supports: The complete route returns to linguistic reality

Abstraction begins from linguistic reality, but a formal model can travel far from the observations that motivated it. Verification secures its internal consequences. Validation tests the proposed correspondence against judgments, distributions, acquisition, variation, and use.

Without that return, the construction remains a possible mathematics rather than an established explanation of language.

MOLR Preface; §6.5